LessonPlanNo	UnitNo	LessonPlanTitle	LessonPlanDescription		
1	1	Introduction to Data Structures	Introduction, need of data structures, classifications (primitive and non-primitive), operations and real-life applications.		
2	1	Review of Arrays	Array concepts, declaration, memory representation and operations.		
3	1	Structures and Unions	Structures, self-referential structures and unions with examples.		
4	1	Pointers and Dynamic Memory	Pointers, malloc(), calloc(), realloc(), free() and dynamic arrays.		
5	1	Algorithm Analysis	Time complexity, space complexity and asymptotic analysis.		
6	1	Performance Analysis	Big-O, Omega and Theta notation with examples.		
7	2	Introduction to Stack	Stack definition, ADT and applications.		
8	2	Stack using Arrays	Array implementation and operations.		
9	2	Dynamic Stack	Stack using dynamic arrays.		
10	2	Applications of Stack	Polish notation, infix to postfix and postfix evaluation.		
11	2	Recursion Basics	Concept of recursion and recursive calls.		
12	2	Recursive Algorithms	Factorial, GCD, Fibonacci and Tower of Hanoi.		
13	2	Introduction to Queue	Queue ADT and operations.		
14	2	Queue Variants	Circular queue, deque and priority queue.		
15	2	Queue Applications	Applications and implementation using arrays.		
16	3	Linked List Basics	Concept, memory representation and node structure.		
17	3	Singly Linked List	Traversal, insertion, deletion and searching.		
18	3	Doubly Linked List	Operations on doubly linked lists.		
19	3	Circular Linked List	Circular linked list operations.		
20	3	Memory Management	Garbage collection and dynamic allocation.		
21	3	Stack using Linked List	Implementation of stack using linked list.		
22	3	Queue using Linked List	Implementation of queue using linked list.		
23	4	Searching Techniques	Linear, binary and interpolation search.		
24	4	Selection Sort	Algorithm and analysis.		
25	4	Insertion Sort	Algorithm and analysis.		
26	4	Bubble Sort	Algorithm and analysis.		
27	4	Quick Sort	Partitioning and recursion.		
28	4	Merge Sort	Divide and conquer approach.		
29	5	Introduction to Trees	Terminologies and binary trees.		
30	5	Binary Tree Representation	Array and linked representation.		
31	5	Tree Traversals	Preorder, inorder and postorder.		
32	5	Binary Search Tree	Insertion, deletion and searching.		
33	5	Expression Trees	Expression evaluation using trees.		
34	5	Balanced Trees	Need and characteristics.		
35	5	Tree Height	Height calculation and applications.		
36	6	AVL Trees	Construction and rotations.		
37	6	AVL Operations	Insertion and deletion.		
38	6	Red Black Trees	Introduction and properties.		
39	6	Red Black Operations	Insertion and deletion overview.		
40	7	Hashing Concepts	Hash tables and hash functions.		
41	7	Static Hashing	Collision handling and applications.		

42	7	Dynamic Hashing	Extendible and dynamic hashing overview.		
43	8	Graph Fundamentals	Terminologies and representations.		
44	8	Graph Traversals	Breadth First Search and Depth First Search.		
45	8	Graph Applications	Real-world applications and revision.		